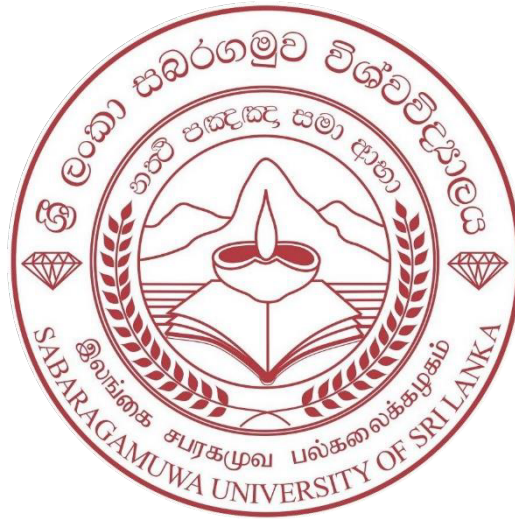




Project in Web Systems & Technologies – SE4106

SkillForge

A gamified educational platform designed to inspire continuous learning and guide users toward optimal knowledge pathways

GROUP NO 17 – SOFTWARE ENGINEERING 22/23

DEPARTMENT OF SOFTWARE ENGINEERING

FACULTY OF COMPUTING

SABARAGAMUWA UNIVERSITY OF SRI LANKA

15th March 2026

Approval of Web Project

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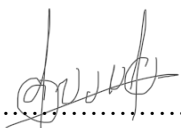
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R. Thidushan - 22CSE0385

W.G.M. Geewinda - 22CSE0386

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Finally, we would like to thank our families and friends for their continuous encouragement, patience, and moral support throughout the development of this project.

GROUP DETAILS AND CONTRIBUTION

Contribution Area	R. Thidushan (Full-stack Developer)	K.G.T.R. Karunajeewa (Full-stack Developer)	W.G.M. Geewinda (Full-stack Developer)
Frontend Development	- Designed admin dashboard layout and workflow validation screens. - Tested UI components and implemented minor UI fixes.	- Created wireframes, user flows and form validation rules. - Assisted with frontend API integration points.	- Developed interactive and responsive web UI screen. - Built the idea of overall UI / UX components of the system.
Backend Development	- Implemented admin-level APIs and role validation logics. - Implemented test endpoints and assisted in bug fixes	- Design database schema and system workflows (ERD, DFD). - Developed core REST APIs authentication, verification logic	Integrated fronted with backend APIs
Cloud and DevOps	- Managed deployment planning and cloud architecture decisions. - Conducted deployment testing and monitored system stability.	- Configured AWS services, hosting, and database. - Analyzed feasibility of cloud services and data storage.	Assisted with environment setup and API testing.
UI / HCI Design	- Coordinated usability goals and emergency interaction constraints. - Conducted usability testing and reported UX issues.	- Designed user skill roadmaps. - Ensured backend responses support fast UX.	UI/UX principles are applied to web interfaces.
Information	- Defined report - Tested misuses	Design verification and implemented flagging, misuse.	Displayed verification

Verification and Validation	• Credibility policy and escalation rules. • Scenarios and false reporting cases.	• Voting and verification scoring logic. • Workflow and validation rules.	• Status and badges in UI.
OpenAPI Services	• Defined API documentation standards and reviewed endpoint structures. • Tested API endpoints and validated responses using API testing tools.	• Designed and implemented RESTful APIs based on OpenAPI specifications. • Generated API documentation and managed request/response validation.	• Integrated frontend components with OpenAPI endpoints. • Handled API requests in the UI and displayed response data effectively.
Notifications and Alerts	• Defined alert prioritization and responder routing logic. • Tested alert delivery and failure scenarios	• Modeled notification triggers and system flow.	• Displayed alert feedback and status updates.
Security and Privacy	• Ensured compliance with privacy policies and access rules. • Conducted security and access testing.	• Defined RBAC rules and data protection requirements. • Implemented JWT authentication and secure APIs.	• Applied frontend access restrictions.
Testing and Quality Assurance	• Reviewed test strategy and milestone validation. • Led functional, integration, and usability testing.	• Prepared test scenarios based on requirements. • Fixed backend issues identified during testing.	• Fixed UI bugs found during testing.
Documentation and Research	• Prepared proposal, reports, and progress documentation. • Maintained test reports and defect logs.	• Created system models, diagrams, and requirement docs. • Documented APIs and backend workflows.	• Documented UI components and usage.

1. INTRODUCTION AND OBJECTIVE OF THE WEB PROJECT

1.1 Introduction

Today, students have access to more information and learning resources than ever before. However, this abundance of choice often creates confusion rather than clarity. Many learners find it difficult to know where to start or how to move forward when trying to develop a new skill. Without a clear structure or guidance, the learning process can quickly become overwhelming and discouraging.

To solve this problem, this project proposes the development of an AI-driven Skill Encouragement Platform, a website designed to help learners follow a clear path from beginner to professional level. The platform provides structured learning plans, task management tools, and motivational support to keep users on track toward their goals.

The platform is built around two key ideas: Micro-Learning and Gamification. Instead of giving learners too much information at once, the platform divides large topics into small, manageable steps called checkpoints. Each checkpoint is further divided into checklists which are given in the format of simple daily tasks that help users to make a steady progress. If a task is not completed on time, the system automatically reschedules it to the next day and provides AI-generated encouragement along with helpful learning materials such as videos, podcasts, and articles.

When learners complete an entire roadmap, they receive a **Golden Batch** and **System Coins** as rewards. During the learning journey, they can also join a **Learner Community** where members ask and answer questions. Asking a question costs coins, while answering earns coins, which encourages active and meaningful participation. In addition, users can use their earned coins to book a **30-minute free online session with an industry expert** in their field of interest. This gives learners a chance to clear doubts, get career guidance, and understand real-world industry expectations, helping connect self-learning with professional mentorship.

The platform also includes a recognition system that tracks how much each user contributes to the community. Based on their activity, users can earn different badges and appear on a leaderboard. These features create a friendly sense of competition, encourage ongoing participation, and reward users who help others learn. Overall, the platform combines artificial intelligence, structured learning, and community engagement to make the learning experience more enjoyable, motivating, and effective for students.

1.2 Objective of the Project

The main goal of the SkillForge platform is to build an AI-powered website that helps learners turn their interest in a particular skill into real, practical ability. The platform will do this by providing clear learning plans, daily task management, smart resource suggestions, a reward system, and a community space where learners can share knowledge with each other.

1. To develop a skill search feature

Users can search for any skill or career they are interested in, and the system will create a personalized learning roadmap to guide them from the beginning of their journey.

2. To create a roadmap-based learning structure

The system breaks down complex skills into smaller steps called checkpoints, giving users a simple and clear path to follow as they grow from beginner to professional level.

3. To build a daily task management system

The system organizes learning activities into small daily tasks, making it easier for users to learn consistently without feeling overwhelmed.

4. To add AI-based learning resource suggestions

The system recommends useful learning materials such as videos, courses, podcasts, and articles that match the user's chosen skill and their current level of progress.

5. To include an automatic task rescheduling feature

If a user does not complete a task on time, the system moves it to the next day and sends an encouraging message to help the user stay motivated and keep going.

6. To introduce a coin-based reward system

The system awards System Coins to users when they complete their daily tasks, making the learning process more engaging and rewarding.

7. To create a digital badge system

The system gives users a digital badge, when they complete checkpoints to fulfil their roadmap. When a user's entire roadmap is finished, they receive a special Golden Badge as the highest form of recognition.

8. To build a community access and knowledge sharing system

The user has access to an exclusive Learner Community. Inside the community, users can ask questions by spending coins and earn coins by answering questions, encouraging everyone to participate and share their knowledge.

9. To add a community leaderboard and recognition system

The system keeps track of how much each member contributes to the community and rewards active users with badges. A leaderboard shows the top contributors to the community.

10. To introduce a coin-based expert session feature

The system allows users to redeem their earned System Coins to access a 30-minute free online session with an industry expert in their field of interest, enabling learners to receive short professional guidance and mentorship.

1.3 Novelty of the SkillForge Application

The SkillForge application introduces several distinctive features that set it apart from existing e-learning and skill development platforms. While most online learning tools simply offer a collection of courses, SkillForge combines AI-driven roadmap generation, structured micro-learning, adaptive task management, and a community-based knowledge exchange system to deliver a personalized, engaging, and practical learning experience.

1. AI-Generated Personalized Learning Roadmaps

SkillForge allows users to search for any skill or career they are interested in, and the system automatically generates a personalized learning roadmap tailored to that specific goal. Unlike traditional platforms that offer fixed course catalogs, SkillForge creates a unique set of checkpoints and learning steps that guide each user on a clear and structured path from beginner to professional level.

2. Micro-Learning Checkpoint Architecture

Rather than presenting learners with large amounts of information at once, SkillForge breaks down complex skill areas into small, manageable checkpoints, each made up of simple checklists which are split into daily tasks. This approach reduces the feeling of being overwhelmed and helps users build their skills gradually and consistently. This structured breakdown of knowledge at the architectural level is not commonly found in traditional e-learning platforms.

3. Adaptive Task Rescheduling with AI Motivational Prompts

When a user fails to complete a task on time, SkillForge does not leave it unresolved. Instead, the system automatically moves the task to the following day and sends an AI-generated motivational message along with relevant learning resources. This feature helps users stay on track and reduces the chance of them giving up on their learning journey.

4. Coin-Based Reward System

SkillForge uses a coin-based reward system to make learning more engaging and motivating. Users earn System Coins when they complete their daily tasks, giving them a sense of achievement at every step. This ongoing reward mechanism encourages users to maintain consistent learning habits throughout their roadmap.

5. Progressive Digital Badges and Golden Badge Recognition

As users' complete checkpoints along their roadmap, they receive digital badges that recognize their progress and achievements. Upon completing an entire roadmap, users are awarded a special Golden Badge, which serves as the highest level of recognition on the platform. These badges can be shared, giving users a sense of pride and accomplishment at every stage of their learning journey.

6. AI-Driven Learning Resource Recommendations

Unlike platforms that offer generic libraries of content, SkillForge uses artificial intelligence to recommend learning materials that are directly relevant to the user's chosen skill and current progress. These recommendations include videos, courses, podcasts, and reading articles, ensuring that users always have access to the most suitable resources at each stage of their roadmap.

7. Exclusive Community Access for Roadmap Completers

One of the most unique features of SkillForge is its exclusive Learner Community. The community is made up of dedicated and knowledgeable learners. Within the community, members can ask questions by spending coins and earn coins by providing answers, creating a self-regulating and high-quality knowledge sharing environment that is not commonly seen in traditional learning platforms.

8. Community Leaderboard and Badge Recognition System

SkillForge tracks each community member's level of participation and contribution. Based on their activity, users earn achievement badges and are featured on a community leaderboard. This recognition system creates a healthy sense of competition, motivates users to keep contributing, and rewards those who actively help others within the community.

9. Coin-Based Access to Industry Expert Sessions

A distinctive feature of SkillForge is the ability for users to redeem System Coins for a 30-minute online session with an industry expert related to their chosen skill domain. While many learning platforms only provide pre-recorded content or peer discussion forums, SkillForge extends learning beyond the platform by giving users direct, short-duration access to real professionals. This feature adds practical value, strengthens learner motivation, and creates a meaningful connection between structured learning and industry insight.

2. ANALYSIS

2.1 Problem and Requirement Analysis

Problem Summary

In the contemporary digital learning landscape, learners are increasingly confronted with information development, while students aspiring to advance from foundational interest to professional competency frequently encounter a significant deficiency in structured pedagogical guidance that limits their capacity to systematically organize and prioritize learning activities. Predominant online learning platforms demonstrate considerable inadequacy in generating personalized, goal-oriented learning roadmaps tailored to specific skill domains and exhibit notable limitations in addressing the behavioral and motivational dimensions of learning, rendering independent learners particularly susceptible to inconsistency, procrastination, and progressive disengagement, further compounded by the pronounced absence of well-structured micro-learning frameworks and robust gamification mechanisms essential for sustaining learner engagement and intrinsic motivation. This project therefore endeavors to systematically address these challenges through the design and implementation of an AI-powered, web-based Skill Encouragement Platform, engineered to deliver structured learning support, adaptive task management, intelligent resource recommendations, and motivational reinforcement mechanisms for learners across diverse skill domains. Furthermore, many learners lack opportunities to interact directly with experienced industry professionals who can provide concise guidance, practical advice, and career-oriented direction. Without access to expert insight, learners may struggle to validate their progress or connect theoretical learning with real industry expectations.

High-Level Goals

- Help learners find a clear path to develop a skill from beginner to professional level.
- Let users search for a skill or career and automatically create a personalized learning roadmap for them.
- Break down complex skills into small checkpoints which are made up of simple checklists. Each checklist is presented as simple daily tasks to make learning easy and manageable.
- Recommend suitable videos, courses, podcasts, and articles based on the user's chosen skill and current progress.
- Automatically reschedule missed tasks to the next day and send encouraging messages to keep users motivated.

- Award System Coins to users when they complete daily tasks to make learning more fun and rewarding.
- Give digital badges when users finish checkpoints and award a special Golden Badge when they complete the full roadmap.
- Give users access to an exclusive Learner Community where they can ask questions by spending coins and earn coins by answering questions.
- Track community contributions and reward active members with badges and a place on the leaderboard.
- Allow users to redeem System Coins for a 30-minute online session with an industry expert in their field of interest.

Stakeholders

- **Learners / Students** — Primary users of the platform who engage with learning roadmaps, complete daily tasks, and accumulate rewards.
- **Platform Administrators** — Personnel responsible for managing platform content, monitoring user activity, and overseeing system operations.
- **Content Providers** — Third-party educators, course creators, and resource publishers
- **AI Recommendation Engine** — The intelligent system component responsible for roadmap generation, resource recommendations, and motivational prompt delivery.
- **System Administrators / DevOps** — Technical personnel responsible for maintaining platform infrastructure, security, and operational performance.
- **Third-Party Service Providers** — External platforms and API services, including video hosting, podcast providers, and cloud infrastructure, that support core platform functionality.

2.2 Feasibility Study

2.2.1 Technical Feasibility

SkillForge is technically feasible because it uses reliable and widely supported technologies for an AI-driven, gamified learning platform. **Next.js 14** manages both the frontend and backend in one project, making development simpler. The **Google Gemini API** powers roadmap generation, motivational messages, and resource recommendations. **MySQL** stores important platform data such as user profiles, tasks, coins, badges, community posts, and leaderboards, while **Prisma** makes database management easier. **Redis** supports fast caching for real-time coin updates and leaderboard performance.

- Next.js 14 supports full-stack development within a single framework, reducing integration complexity.
- Google Gemini API enables AI-driven roadmap generation, motivational prompts, and resource recommendations.
- MySQL and Prisma provide a stable and well-managed relational database solution.
- Redis ensures fast caching for coin transactions and leaderboard updates.
- NextAuth.js and Google OAuth provide secure and easy-to-integrate user authentication.
- Vercel and Railway offer reliable and cost-effective deployment environments.
- node-cron enables automated daily task rescheduling without additional infrastructure.
- All selected technologies are open-source, well-documented, and actively maintained.
- The platform can support expert session booking through additional scheduling and session management functionality.
- Online expert sessions can be integrated using standard video meeting technologies such as Google Meet or Zoom links.
- The existing coin transaction system can be extended to support coin redemption for expert session access.

The project is **Technically Feasible** because all required technologies are mature and collectively capable of delivering the full feature set of SkillForge without significant technical risk. The expert session feature is technically feasible because it can be implemented using existing web technologies, scheduling logic, coin transaction handling, and third-party meeting integration tools.

2.2.2 Operational Feasibility

The SkillForge platform is operationally feasible as it is designed to fit naturally into the daily learning routines of students and independent learners without requiring any specialized technical knowledge. The user interface is built with Tailwind CSS, Shadcn/UI, and Framer Motion to provide a clean and intuitive experience that minimizes the learning curve for new users. The platform supports clearly defined roles for learners and administrators, ensuring smooth system operation. The automated task rescheduling system powered by node-cron and the Google Gemini API operate autonomously, reducing the need for manual administrator involvement. The coin-based community system is self-regulating, maintaining discussion quality without external moderation, while the leaderboard and badge system automatically rewards active contributors.

- Intuitive user interface minimizes training requirements for learners and administrators.
- Clearly defined role separation ensures smooth and efficient platform operation.
- Google Gemini API autonomously delivers roadmaps, motivational prompts, and resource recommendations.
- node-cron automates daily task rescheduling without manual intervention.
- The coin-based community system is self-regulating and maintains high-quality participation.
- The exclusive Learner Community ensures a focused and knowledgeable membership base.
- The platform is accessible via standard web browsers without any special hardware or software.
- The platform supports scalable onboarding of new learners across diverse skill domains.
- The expert session feature provides a simple and user-friendly way for learners to access professional guidance using their earned coins.
- Administrators can manage expert availability, session approvals, and booking confirmations through the platform.
- Short 30-minute sessions reduce operational complexity while still offering meaningful learner support.

The project is **Operationally Feasible** because all features are designed to run with minimal manual intervention, allowing learners and administrators to use the platform effectively and efficiently.

2.2.3 Economic Feasibility

The SkillForge platform is economically feasible due to its strategic use of open-source frameworks, free-tier cloud services, and cost-effective API integrations that keep development and operational costs low. The primary stack, including Next.js, Node.js, MySQL, Prisma, and Redis, is either fully open-source or available at no licensing cost. Vercel and Railway provide affordable deployment infrastructure with costs that scale proportionally to platform growth. The Google Gemini API delivers significant value through automated roadmap generation and content delivery, removing the need for a dedicated content team. GitHub and GitHub Projects provide free version control and project management tools throughout the development lifecycle.

- Next.js, Node.js, Prisma, and MySQL are open-source with no licensing fees.
- Vercel and Railway offer free and low-cost deployment tiers suitable for initial launch.
- Redis free tier is sufficient for caching coin transactions and leaderboard data.
- Google Gemini API costs are usage-based, allowing controlled and predictable spending.
- NextAuth.js provides free, open-source authentication, eliminating third-party subscription costs.
- GitHub and GitHub Projects provide free version control and project management tools.
- node-cron is a free scheduling library, removing the need for paid job queue services.
- Automation of AI, rescheduling, and community features reduces long-term staffing costs.
- No specialized hardware is required, reducing capital expenditure significantly.
- The 10-minute session model keeps mentorship interaction brief and cost-efficient, making the feature practical for academic project scope.
- Session booking can initially rely on free meeting platforms and limited expert participation, reducing additional operational expenses.

The project is **Economically Feasible** because the total development and operational costs are low relative to the significant educational value and community-driven benefits delivered to learners.

2.3 Data Flow Diagrams (DFD)

2.3.1 Context Diagram

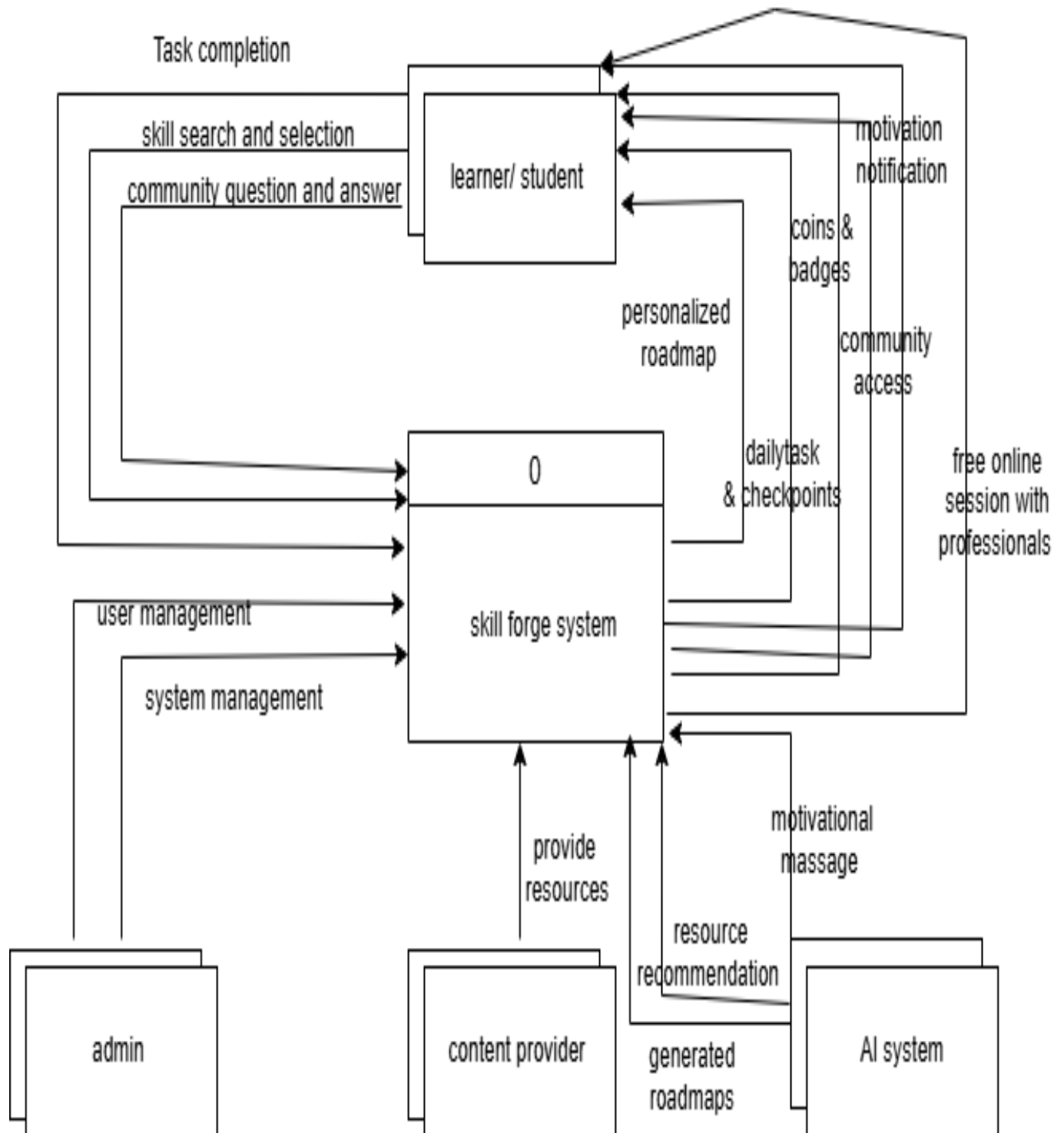


Figure 01: Context Diagram of the SkillForge system

2.3.2 Level 1 Data Flow Diagram

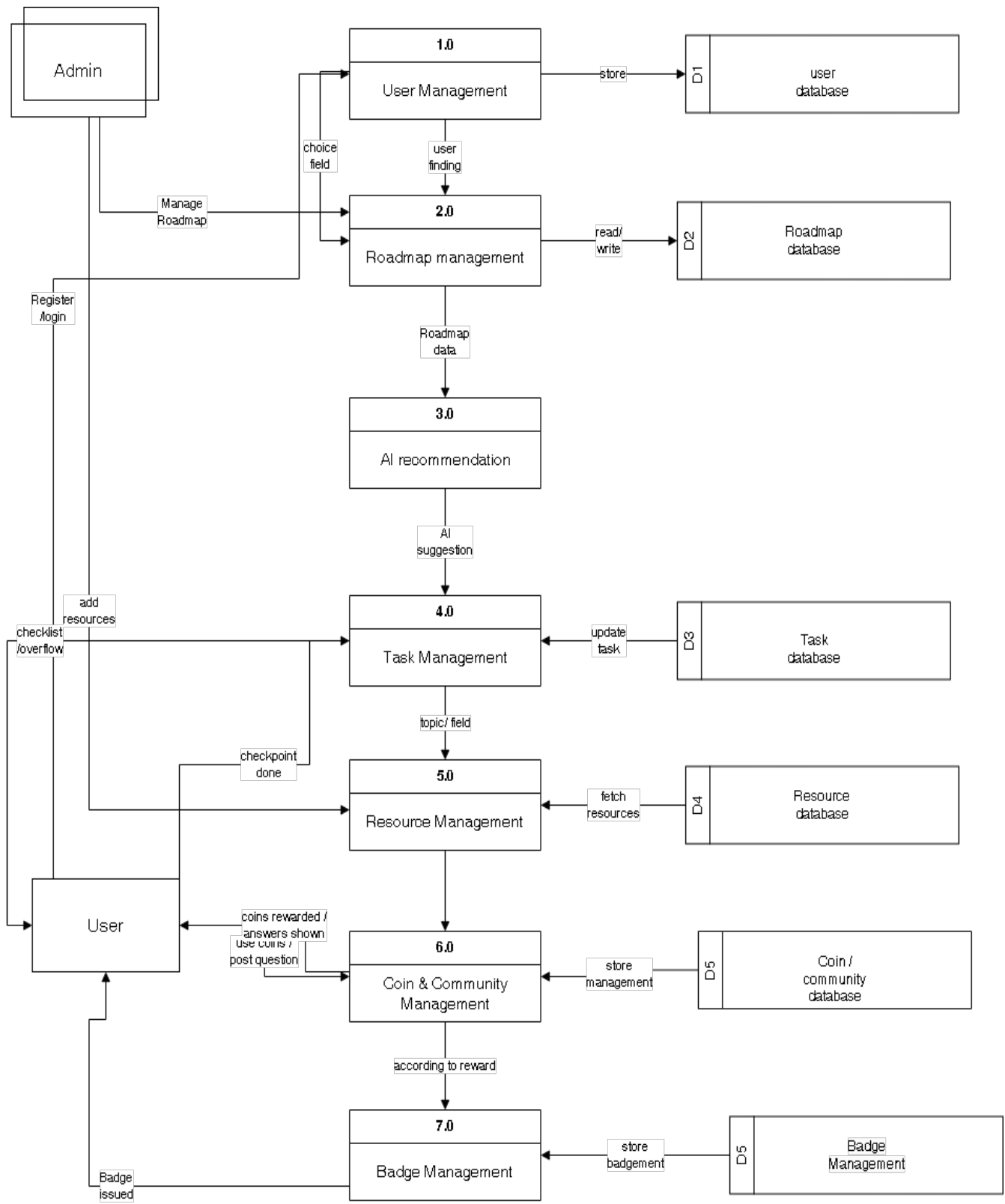


Figure 02: Level 1 Data Flow Diagram (DFD)

2.4 Entity Relationship (ER) Diagram

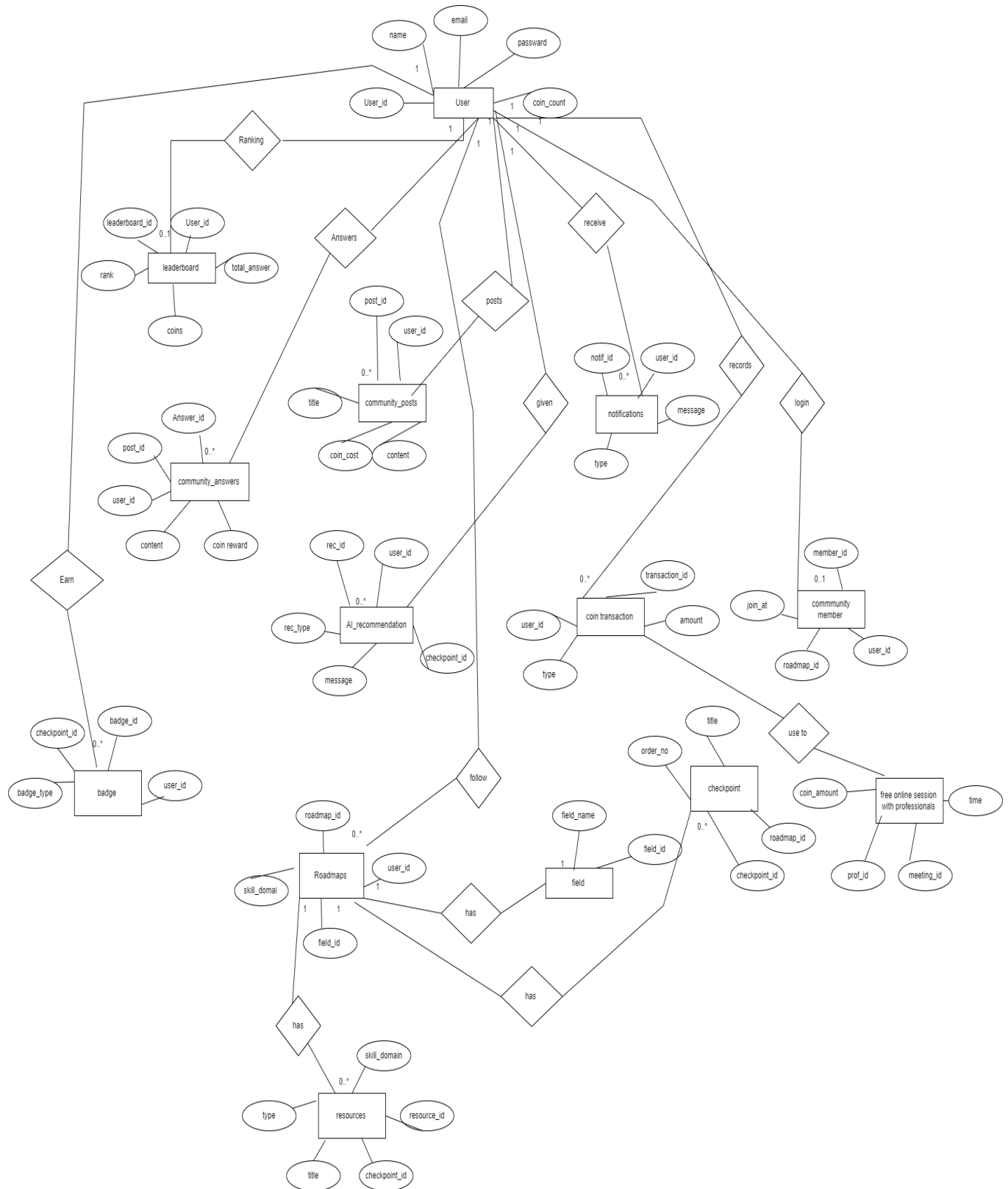


Figure 03: Entity - Relationship (ER) Diagram

2.5 Use Case Diagram

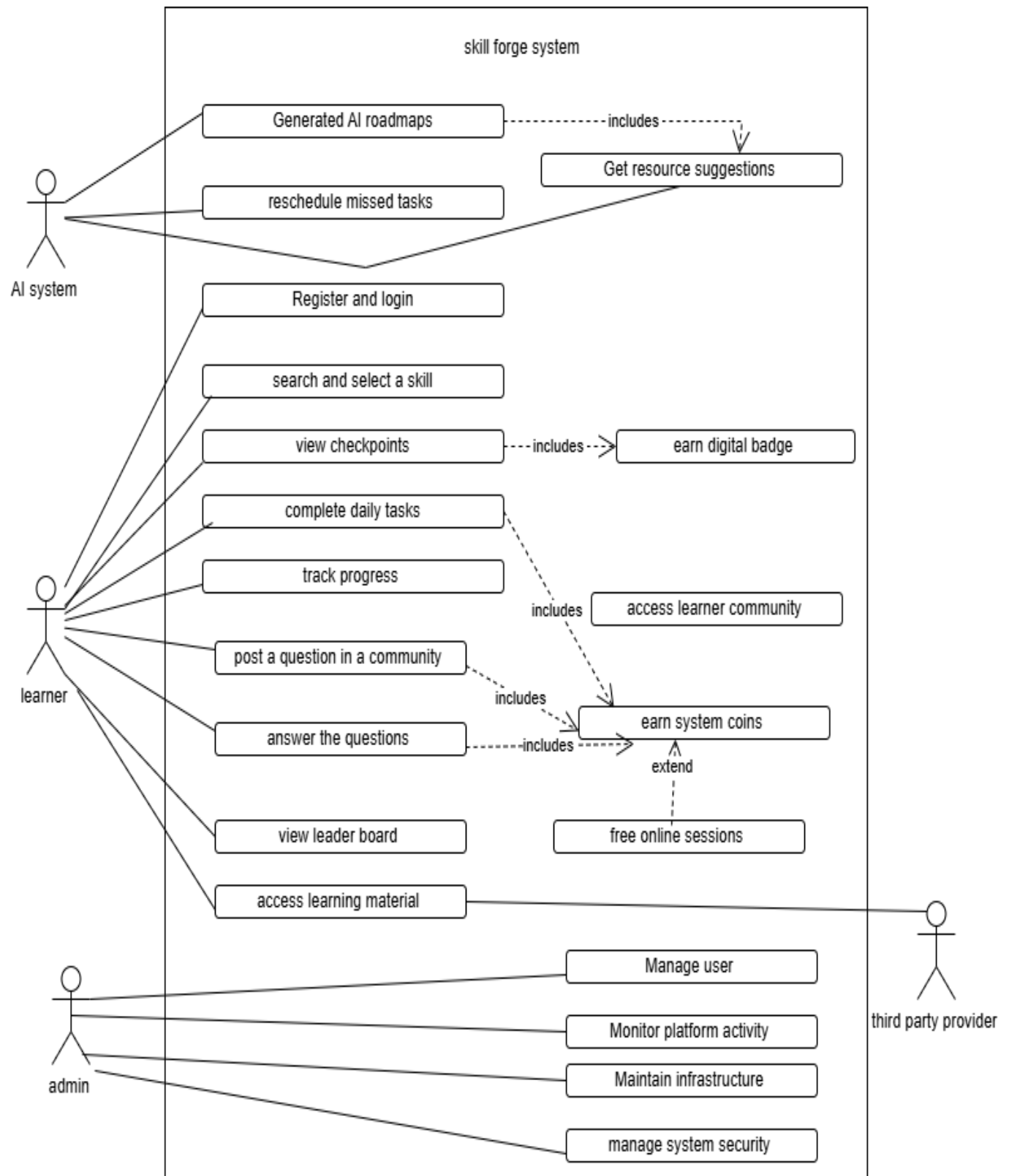


Figure 04: Use Case Diagram of the SkillForge system

3. HARDWARE AND SOFTWARE REQUIREMENT

3.1 Hardware Requirements

Hardware Component	Minimum Specification
Development Workstation	Processor: Intel Core i5 (10th Gen) / RAM: 16GB DDR4 / Storage: 256GB SSD / OS: Windows 10 / macOS 12 / Ubuntu 20.04 LTS
Storage	256 GB SSD minimum, 512 GB SSD recommended, to support the application source code, MySQL database storage, cached learning resources, user progress records, community discussion data, coin transaction logs, and system backups. Cloud: Managed MySQL hosting / cloud database service for relational data storage, with AWS S3 or Cloundinary free tier for badge and media file storage.
Network Connectivity	Minimum: 10 Mbps broadband internet connection / Recommended: 25 Mbps or above for stable AI API calls, Socket.io real-time communication, and cloud service integration
Display	Minimum: 1366 x 768 resolution (HD) / Recommended: 1920 x 1080 (Full HD) / Size: 13 inches or above for effective UI development and testing of roadmap and gamification interfaces
Mobile / Tablet Device	OS: Android 10.0 or iOS 14.0 and above / RAM: 3GB minimum / Screen Resolution: 720 x 1280 (HD) or above / Browser: Latest version of Chrome or Safari for responsive UI testing of the SkillForge web platform

3.2 Software Requirement

Category	Technology and Tools	Purpose
Frontend Framework	Next.js 14 (App Router)	Core frontend framework providing server-side rendering, static site generation, and full-stack capabilities for the SkillForge web application
UI Component Library	Tailwind CSS, Framer Motion, Shadcn/UI, Lottie	Responsive UI styling, gamification animations, badge rendering, coin reward effects, and smooth transitions across roadmap and community pages
Backend Framework	Node.js, Express.js, Next.js API Routes, node-cro	Server-side logic, RESTful API development, automated daily task rescheduling, and job scheduling for missed task detection and motivational prompt delivery
Database Management	MySQL	Primary relational database for storing user profiles, roadmaps, checkpoints, daily tasks, coin balances, badge records, community posts, and leaderboard data in a structured and well-organized table format
ORM	Prisma	Schema modeling, data validation, database migration management, and interaction layer between Node.js and MySQL
Caching	Redis	Real-time caching of coin transaction data, community activity tracking, and leaderboard ranking updates for improved performance
Authentication	NextAuth.js, Google OAuth, bcrypt.js, JWT	Secure user authentication, social login via Google, password hashing, and session token management
Version Control	Git, GitHub	Source code version control, collaborative development, branch management, and code review
Coin and Reward System	Custom Business Logic via Node.js	Managing coin earning and spending transactions, tracking daily task completion for coin rewards, handling coin deduction for community questions, and coin addition for community answers

Community Module	Next.js API Routes, MySQL	Handling community access control for roadmap completers, managing question and answer posts, processing coin-based transactions between members, and tracking member participation
Badge and Leaderboard System	Custom Logic via Node.js, MySQL	Awarding digital badges upon checkpoint completion, granting the Golden Badge upon full roadmap completion, tracking community contributions, and updating leaderboard rankings based on member activity
Project Management	GitHub Projects	Sprint planning, task tracking, milestone management, and team collaboration throughout the development lifecycle
Session Scheduling and Meeting Integration	Next.js API Routes, Node.js, Google Meet / Zoom Links	Handling expert booking requests, session confirmations, scheduling logic, and online meeting link generation
Frontend Deployment	Vercel	Primary deployment platform optimized for Next.js applications, providing CI/CD integration, serverless functions, and global CDN delivery
Backend Deployment	Railway / Render, Docker, GitHub Actions	Hosting the Node.js and Socket.io microservice, containerization of backend services, and automated CI/CD pipeline for testing and deployment
Integrated Development Environment	Visual Studio Code (VS Code)	Primary code editor with extensions for Next.js, Tailwind CSS, ESLint, Prettier, and MongoDB for efficient development workflow
Automated Testing	Postman, Jest	Unit and integration testing for backend logic and coin calculation, end-to-end testing for the timer and coin collection flow, and API endpoint testing
Operating System	Windows 10 / macOS 12 / Ubuntu 20.04 LTS	Development environment operating systems compatible with all selected tools, frameworks, and cloud service integrations

4. TABLES, STRUCTURE, NUMBER OF MODULES AND DATA STRUCTURES

4.1 Number of Modules and Module Descriptions

The proposed SkillForge system is architecturally divided into eleven functional modules. Each of these modules is responsible for a major area of operation of the platform. These modules work together to support to perform user onboarding, AI-driven roadmap generation, checkpoint and task management, gamification, learning resource delivery and achievement tracking. The functional modules are described in the following table.

Module No.	Module Name	Module Description
1	User Management Module	Manages processes like user registration, login, profile management, user authentication and account settings
2	Skill Search and Selection Module	Enables users to search for their desired skill or learning field and select it as their learning path
3	AI Roadmap Generation Module	Generates a structured roadmap to achieve the selected skill using AI-based recommendations.
4	Checkpoint Management Module	Organizes the roadmap into effective checkpoints and monitors the status of completion of each checkpoint.
5	Daily Task Management Module	Breaks each checkpoint into simple daily tasks and organizes them in a manageable sequence to support consistent learning progress
6	Adaptive Task Rescheduling Module	Automatically reschedules incomplete tasks to the following day and delivers AI-generated motivational messages to encourage users to continue
7	Progress Tracking Module	Tracks user progress across roadmaps, checkpoints, and daily tasks to provide a clear view of overall learning advancement
8	AI Resource Recommendation Module	Recommends relevant learning materials such as videos, courses, podcasts, and articles based on the user's chosen skill and current progress.
9	Coin-Based Reward and Redemption Module	Manages the earning, spending, and redemption of System Coins, including rewards for daily task completion, community transactions, and expert session bookings.

10	Badge and Achievement Module	Awards digital badges upon checkpoint completion and grants a special Golden Badge when a user successfully completes an entire roadmap
11	Community Access Module	Grants access for users to the Learner Community for managing membership control and community participation.
12	Community Knowledge Exchange Module	Manages the posting of questions and answers within the community, processing coin deductions for questions and coin rewards for answers
13	Community Leaderboard and Recognition Module	Tracks each community member's contributions and engagement, awards achievement badges, and maintains the community leaderboard rankings
14	Expert Session Booking Module	Enables users to redeem System Coins to book a 30-minute online session with an industry expert in their selected field of interest, while managing booking requests, session scheduling, confirmations, and meeting access links.

4.2 Database Structures (Key Entities and Attributes)

The proposed SkillForge system requires a well-structured relational database to manage users, selected skills, AI-generated roadmaps, checkpoints, daily tasks, coin transactions, community posts, badges, leaderboard data, and learning resources effectively and efficiently. The database is designed to ensure consistency, scalability, and reliable retrieval of user progress data while supporting personalized recommendations and community interactions. Each table is responsible for storing a specific category of system data while maintaining clear relationships with other tables through primary and foreign keys.

The following table describes the main database tables and their corresponding purposes within the proposed SkillForge system.

Collection / Entity	Key Attributes	Description
users	user_id (PK), name, email, password_hash, role, auth_provider, created_at	Stores credentials, authentication provider details, and role assignments for all system users
profiles	profile_id (PK), user_id (FK), preferences, coin_balance, star_balance, active_roadmap_ids, created_at	Holds extended profile data, reward balances, and active roadmap references specific to each learner

roadmaps	roadmap_id (PK), user_id (FK), skill_domain, title, total_checkpoints, status, created_at	Stores AI-generated learning roadmap records associated with each user's selected skill or career domain
checkpoints	checkpoint_id (PK), roadmap_id (FK), title, order_index, status, stars_awarded, badge_id (FK)	Defines each sequential checkpoint within a roadmap, including completion status and associated reward references
daily_tasks	task_id (PK), checkpoint_id (FK), user_id (FK), title, status, scheduled_date, rescheduled_date, created_at	Stores individual daily tasks assigned to each checkpoint, including scheduling and rescheduling details for incomplete tasks
coin_transactions	transaction_id (PK), user_id (FK), type, amount, reference_id, description, created_at	Records all coin earning, spending, and redemption transactions, including task completions, community question deductions, answer rewards, and expert session bookings.
notifications	notif_id (PK), user_id (FK), message, type, sent_at, is_read	Archives all system-generated notifications, session reminders, booking confirmations, and cancellation updates are dispatched to authenticated users
resources	resource_id (PK), checkpoint_id (FK), title, type, access_level, url, skill_domain	Contains AI-recommended learning resources linked to specific checkpoints categorized by format type
badges	badge_id (PK), user_id (FK), checkpoint_id (FK), badge_type, is_golden, awarded_at	Records digital achievement badges awarded to users upon checkpoint completion and full roadmap completion, with a flag identifying Golden Badge awards
community_members	member_id (PK), user_id (FK), roadmap_id (FK), joined_at	Stores membership records for the exclusive Learner Community, restricted to users who have completed a roadmap
community_answers	answer_id (PK), post_id (FK), user_id (FK), content, coin_reward, created_at	Stores answers submitted by community members in response to posted questions, including the coin reward granted for each answer

leaderboard	leaderboard_id (PK), user_id (FK), total_answers, coins_earned, badges_count, rank, updated_at	Tracks each community member's overall contribution score, badge count, and leaderboard ranking based on their participation
industry_experts	expert_id (PK), name, email, field_of_expertise, bio, availability_status, meeting_link, created_at	Stores expert profile details, specialization, availability, and session access information.
expert_sessions	session_id (PK), user_id (FK), expert_id (FK), skill_domain, scheduled_date, start_time, duration_minutes, session_status, coin_cost, meeting_link, created_at	Stores booked 30-minute expert sessions, including learner, expert, time, status, and coin redemption details.
expert_session_requests	request_id (PK), user_id (FK), field_of_interest, preferred_date, preferred_time, request_status, created_at	Stores user requests for expert sessions before final expert assignment and booking confirmation.

5. PROPOSED SYSTEM

5.1 Functional Requirements

The functional requirements specify the behaviors and capabilities that the SkillForge system must deliver upon implementation. Each requirement is directly linked to the project objectives defined in Section 1.2.

Administrator

- The system shall provide authenticated administrators with the ability to manage all registered user accounts, including updating account details and deactivating accounts when necessary.
- The system shall allow authenticated administrators to review, edit, or remove AI-generated learning roadmaps that contain inaccurate or inappropriate content.
- The system shall enable authenticated administrators to monitor all active and completed learning roadmaps across the platform through a centralized management dashboard.
- The system shall provide authenticated administrators with access to a real-time analytics dashboard displaying platform-wide performance data, including daily active users, roadmap completion rates, and community activity statistics.
- The system shall allow authenticated administrators to manage the badge and coin reward system, including adjusting coin values assigned to specific tasks and community activities.
- The system shall allow authenticated administrators to manage registered industry experts, including adding, updating, and removing expert profiles and their fields of expertise.
- The system shall allow authenticated administrators to manage expert session availability, booking requests, and confirmations.
- The system shall enable authenticated administrators to monitor all expert session bookings and related coin redemption activities.

User

- The system shall allow users to register and securely log in using their email and password or via Google OAuth.
- The system shall allow authenticated users to search for any skill or career they are interested in and receive an AI-generated personalized learning roadmap tailored to that goal.

- The system shall present each active roadmap as a series of sequential checkpoints, each containing a set of simple daily tasks for the user to complete.
- The system shall allow authenticated users to mark daily tasks as complete and view their overall progress across checkpoints and the full roadmap on their personal dashboard.
- The system shall provide authenticated users with AI-recommended learning materials, including videos, courses, podcasts, and articles, relevant to their chosen skill and current checkpoint.
- The system shall automatically reschedule any incomplete daily tasks to the following day and send an AI-generated motivational message to encourage the user to continue.
- The system shall award System Coins to authenticated users each time they successfully complete a daily task, with the coin balance visible on the user's profile dashboard.
- The system shall issue digital achievement badges to authenticated users upon the completion of each checkpoint, and award a special Golden Badge upon the successful completion of an entire roadmap.
- The system shall allow authenticated users to manage their personal profile, including preferences, active roadmaps, and coin balance.
- The system shall send automated notifications and reminders to authenticated users about pending daily tasks and upcoming checkpoint deadlines.
- The system shall grant authenticated users access to the exclusive Learner Community.
- The system shall allow community members to post questions by spending coins and earn coins by providing answers to other members' questions.
- The system shall display a community leaderboard that ranks members based on their participation, answers provided, and coins earned within the community.
- The system shall award community achievement badges to users based on their level of contribution and engagement within the Learner Community.
- The system shall allow authenticated users to view available industry experts based on their selected skill domain or field of interest.
- The system shall allow authenticated users to redeem System Coins to book a 30-minute online session with an industry expert.

- The system shall provide authenticated users with session confirmation details, including date, time, and online meeting access link.
- The system shall send automated reminders to users before the scheduled expert session begins.

AI Roadmap Generation System

- The system shall automatically generate a structured and personalized learning roadmap when a user searches for a skill or career, breaking it down into sequential checkpoints and daily tasks.
- The system shall ensure that each generated roadmap is uniquely built based on the user's specific input and learning goal.
- The system shall continuously recommend relevant learning resources at each checkpoint stage, matched to the user's chosen skill and current progress.

Task Management System

- The system shall organize all roadmap content into simple daily tasks and display them in a clear and time-bound task view for each authenticated user.
- The system shall track and update task completion status in real time, immediately reflecting all changes on the user's roadmap view and progress dashboard.
- The system shall automatically move the user to the next checkpoint once all tasks in the current checkpoint have been completed.

Coin and Community System

- The system shall automatically calculate and credit the correct number of System Coins to a user's account upon verified completion of a daily task.
- The system shall deduct coins from a user's balance when they post a question in the community and add coins to a user's balance when they submit an accepted answer.
- The system shall track all coin transactions and maintain an accurate and up-to-date coin balance for every user.

Badge and Leaderboard System

- The system shall award digital badges to users upon the completion of each checkpoint and grant the Golden Badge exclusively upon the completion of an entire roadmap.
- The system shall track each community member's contributions and automatically update their badge collection and leaderboard ranking based on their activity.
- The system shall maintain and publicly display a community leaderboard showing top contributors ranked by their participation, answers, and coins earned.

Expert Session System

- The system shall allow eligible users to request a 30-minute expert session using their available System Coins.
- The system shall verify that the user has sufficient System Coins before confirming the booking.
- The system shall deduct the required number of coins from the user's balance once the booking is confirmed.
- The system shall match or assign an industry expert based on the user's selected field of interest.
- The system shall store expert session booking details and maintain the booking status as pending, confirmed, completed, or cancelled.
- The system shall generate or attach an online meeting link for each confirmed session.
- The system shall notify both the learner and the assigned expert about the scheduled session.

5.2 Non-Functional Requirements

The non-functional requirements define the quality standards and operational boundaries within which the SkillForge system must operate.

- **Performance:** All system pages and AI-generated roadmap responses shall load within three seconds under normal network conditions to provide users with a smooth and uninterrupted learning experience. Expert session booking and confirmation actions shall be processed within a reasonable response time under normal operating conditions.
- **Security:** All user passwords shall be stored in an encrypted format. All API endpoints shall require JWT-based authentication, and HTTPS shall be enforced across the platform to protect user data, coin balances, and community activity. Expert meeting links, booking records, and user session information shall only be accessible to authorized users and administrators.

- **Usability:** The user interface shall be fully responsive and function consistently across desktop, tablet, and mobile devices, allowing learners to access their roadmaps, daily tasks, and the community from any device.
- **Reliability:** The system shall handle errors gracefully by displaying clear and helpful messages to users without revealing internal system details, while ensuring that task records, coin balances, and community data remain accurate at all times. The system shall prevent double-booking of expert sessions and ensure that booking, cancellation, and coin deduction records remain accurate and synchronized.
- **Scalability:** The system architecture shall be designed to accommodate a growing number of users, additional skill domains, expanded community activity, and new features without requiring major changes to the existing codebase.
- **Maintainability:** All source code shall follow clean code standards, include clear comments and documentation, and be managed using GitHub with meaningful commit messages to support future development and maintenance.
- **Compatibility:** The system shall function correctly on all current major web browsers, including Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari.
- **Availability:** The system shall remain accessible during peak usage periods, with fallback mechanisms in place to maintain core learning, progress tracking, community features and session scheduling data in the event of a partial system failure.

5.3 Development Methodology

SkillForge will be developed using an Agile software development methodology, structured around iterative two-week sprint cycles. This approach allows for continuous feedback, early identification of technical issues, and flexible response to changing requirements throughout the project lifecycle. Each sprint will include planning, incremental development, internal review, and retrospective activities.

The project is organized into five sequential phases as follows:

- **Phase 1 — Initiation and Planning (Weeks 1–3):** Topic selection, stakeholder analysis, literature review, feasibility assessment, and preparation of the project proposal, including identification of AI tooling requirements, gamification design principles, and community system design.
- **Phase 2 — System Design (Weeks 4–5):** Entity-relationship diagram design, data flow diagram construction, user interface wireframing, finalization of the technology stack, and definition of the AI roadmap generation workflow, coin reward system, and community access architecture.
- **Phase 3 — Core Module Development (Weeks 6–10):** Incremental implementation of user authentication, profile management, AI-powered roadmap generation, checkpoint and daily task management, automatic task rescheduling, and the coin-based reward system.
- **Phase 4 — Advanced Feature Development (Weeks 11–13):** Implementation of the digital badge and Golden Badge award system, coin-based question and answer system, community leaderboard and badge recognition system, AI-generated motivational prompt delivery, integrated learning resource recommendation module, and the coin-based expert session booking and scheduling feature.
- **Phase 5 — Testing, Deployment, and Documentation (Weeks 13–15):** Execution of unit, integration, and user acceptance testing; defect resolution; cloud deployment via Vercel and Railway; preparation of the final project report; and presentation rehearsal.

6. MODULE SPLIT-UP AND GANTT CHART

6.1 Module Allocation

NO	Module	Responsible Member (Index No.)	Planned Weeks
1	User Authentication and Authorization	K.G.T.R. Karunajeewa (22CSE0377)	6-7
2	User Profile Management	K.G.T.R. Karunajeewa (22CSE0377)	7-8
3	AI Roadmap Generation	R. Thidushan (22CSE0385)	7-9
4	Daily Task Management and Progress Tracking	K.G.T.R. Karunajeewa (22CSE0377)	8-10
5	Adaptive Task Rescheduling and Motivational Prompts	R. Thidushan (22CSE0385)	9-10
6	Coin-Based Reward System	W.G.M. Geewinda (22CSE0386)	10-11
7	Digital Badge and Achievement System	W.G.M. Geewinda (22CSE0386)	11-12
8	AI Learning Resource Recommendation	R. Thidushan (22CSE0385)	11-12
9	Community Leaderboard and Recognition System	K.G.T.R. Karunajeewa (22CSE0377)	11-13
10	Community Access and Knowledge Exchange System	W.G.M. Geewinda (22CSE0386)	11-13
11	Expert Session Booking and Scheduling System	R. Thidushan (22CSE0385)	12-13
12	Notifications System	W.G.M. Geewinda (22CSE0386)	13

6.2 Tentative Work Plan (Gantt Chart)

The Gantt chart presented below illustrates the planned schedule of development activities across the 15-week project semester. Shaded cells indicate weeks during which a given activity is anticipated to be in progress.

Activity	Duration In Weeks														
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Project Planning and Proposal Preparation	■	■	■												
Feasibility Study and Literature Review	■	■	■												
System and Database Design				■	■										
User Interface Wireframing				■	■										
Authentication Module Development						■	■								
User Profile Management Module							■	■							
AI Roadmap Generation Module							■	■	■						
Daily Task Management and Progress Tracking								■	■	■					
Adaptive Rescheduling and Motivational Prompts									■	■					
Coin-Based Reward System Module										■	■				
Digital Badge and Achievement System											■	■			
AI Learning Resource Recommendation Module											■	■			
Community Access and Knowledge Exchange											■	■	■		
Community Leaderboard and Recognition System											■	■	■		
Expert Session Booking and Scheduling Module												■	■		
Notifications System													■		
Integration Testing and QA													■	■	
Cloud Deployment														■	■
Final Report Documentation													■	■	■
Final Presentation Preparation														■	■

7. COST ANALYSIS

The following table presents a comprehensive estimate of the anticipated expenditure associated with the development and deployment of SkillForge. The cost structure reflects a deliberate selection of open-source and free-tier tools, with the objective of minimizing financial outlay whilst maintaining a professional-grade development and deployment environment.

No.	Item Description	Estimated Cost (LKR)	Remarks
1	Domain Name Registration (one year)	3500.00	Optional; required for production deployment
2	Frontend Cloud Hosting – Vercel Free Tier	0.00	Sufficient for academic demonstration purposes
3	Backend Cloud Hosting – Railway Free Tier	0.00	Free plan adequate for project scope
4	Database Hosting – Managed Railway MySQL cloud hosting via a free-tier	0.00	Relational database hosting for user data, roadmaps, checkpoints, community questions, answers, and coin transaction records.
5	Cache Hosting – Upstash Redis Free Tier	0.00	Free tier sufficient for timer session caching
6	AI API – OpenAI GPT-4o (Development Usage)	2500.00	Estimated cost for API calls during development and testing phases
7	Media Storage – Cloudinary Free Tier	0.00	Free tier adequate for badge and resource media storage
8	Integrated Development Environment – Visual Studio Code	0.00	Open-source; no license cost
9	Version Control – GitHub	0.00	Free for student accounts
10	Project Management – GitHub Projects	0.00	Free plan used for sprint tracking
11	API Testing – Postman	0.00	Free plan adequate for project scope
12	Project Proposal and Final Report Printing and Binding	2000.00	Estimated cost for two bound copies
13	Miscellaneous and Contingency	1500.00	Provision for unforeseen minor expenses
14	Expert Session Integration / Meeting Platform	0.00	Free meeting tools or manually generated meeting links sufficient for academic prototype
	TOTAL ESTIMATED COST (LKR)	9500.00	

8. REFERENCES

All references cited in this proposal conform to the IEEE citation style. Each reference listed below corresponds to a citation appearing within the body of the document.

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